

CURRICULUM VITAE

Jin Hyun Park

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EDUCATION

Texas A&M University, College Station, Texas, US
Ph.D. in Computer Science and Engineering
Supervisor: Dr. Yoonsuck Choe

Aug 2022 – Present

University of Bristol, Bristol, England, UK
B.Sc. in Computer Science with First Class Honors
Supervisor: Dr. Rui Ponte Costa

September 2017 – July 2021

RESEARCH INTEREST

Machine Learning, Artificial Intelligence, Computational Neuroscience, and Natural Language Processing

EXPERIENCES

Physical AI PhD Fellow
Doosan Corporation, South Korea

Feb 2026 – Present

Graduate Research and Teaching Assistant | Python, Pytorch, Java
Brain Networks Lab, Texas A&M University, College Station, Texas
[Research] Supervisor: Dr. Yoonsuck Choe

Aug 2022 – Present

- (1) Biologically motivated neural networks, (2) Smart AI floor-heating thermostat, and (3) Evolution of predictive behavior
- Teaching assistant for CSCE 420/625 Artificial Intelligence

Machine Learning Engineer Intern | Python, Pytorch
Machine Learning Team, Tilda Corp., South Korea

May 2023 – Aug 2023 & July 2025 – Aug 2025

- (2023) Machine learning solutions via a neural-network-based genetic algorithm. Implementation details are available at: <https://bit.ly/4hZ2FpO>
- (2025) Meta-heuristic and neural methods for vehicle routing, multi-agent path-finding, and time series forecasting

Machine Learning Engineer Intern | Python, Pytorch
AI Service Part, Digital Innovation Team, Samsung Life Insurance, South Korea

July 2024 – Aug 2024

- Automatic PDF parsing system and search engine development (including Gen-AI and RAG) for insurance documents and policy retrieval (Prototype: <https://shorturl.at/fkj5V>)

Machine Learning Engineer Intern | Python, Pytorch, SQL, and Google BigQuery
AI Prediction Tech Team, Data Science Lab, LG Telecom (a.k.a. LG Uplus), South Korea

May 2024 – July 2024

- Machine learning models (e.g., ensemble modeling and transformers) for identifying roaming outbound travelers.
- Big data generation, pre-processing, and visualization (SQL and Google BigQuery).

Research Assistant | Python, Pytorch
Multimodal Language & Cognition Lab (MLCL), Kyungpook National University, South Korea

Jan 2022 – July 2022

- Reinforcement learning seminars for lab students, code samples, and lecture notes (available at: <https://bit.ly/3Qj1m7u>. Currently 20 stars on GitHub).
- Traditional NLP algorithms and LLM research

Business Strategy Intern
Commercial Vehicle Business Innovation Team, Hyundai Motors Company, South Korea

Oct 2021 – Dec 2021

- Business modeling (short to long-term strategies and plans), with a particular focus on FCEV, BEV, and commercial vehicles.

ACADEMIC ACTIVITIES & TALKS

Reviewer / Program Committee

AAAI 24, 25, 26 / AAAI-AIA 26 / AAAI-Demo 26 / IEEE TCDS 25 / Neural IPS Workshop (NeurReps) 26

Part-time Lecturer (Lectures on Large Language Models)

Yonsei University 2025

PAPERS (W: Workshop, J: Journal, C: Conference, P: Preprint, *: Equal Contribution)

- [W1] Alex Alvarez* and **Jin Hyun Park***, Yoonsuck Choe., *Saliency Thresholds in Neural Code and its Relation to the Power-Law, Gaussian, and Lambert W Function.*, NeurIPS Workshop on Symmetry and Geometry in Neural Representations (NeurReps), 2025., [<https://openreview.net/forum?id=cjo2oadaRz>]
- [J1] William Kang, Chris Anand, **Jin Hyun Park**, Yoonsuck Choe., *Evolutionary Factors Contributing to the Emergence of Prediction.*, Adaptive Behavior, 2025., [<https://journals.sagepub.com/doi/10.1177/10597123251364746>]
- [C1] Seyyed Ali Ayati* and **Jin Hyun Park***, Yichen Cai, Marcus Botacin, *Making Acoustic Side-Channel Attacks on Noisy Keyboards Viable with LLM-Assisted Spectrograms' "Typo" Correction.*, 19th USENIX WOOT Conference on Offensive Technologies, 2025., [<https://arxiv.org/abs/2504.11622>]
- [C2] Soha Lee, Soojin Lee, Heesung Yang, Hyunju Song, Hyunji Lee, Jinsan An, Jeongwan Shin, Jin Hyun Park, Jun Lee, Hyeyoung Park, Kilim Nam., *KoNeoBench: A Curated Evaluation Dataset for LLM Understanding of Korean Neologisms.*, Findings of EMNLP, 2026
- [P1] **Jin Hyun Park**, Cheng Zhang, Yoonsuck Choe., *Incorporating Visual Cortical Lateral Connection Properties into CNN: Recurrent Activation and Excitatory-Inhibitory Separation.*, Arxiv Preprint., 2025., [<https://www.arxiv.org/abs/2509.15460>]
- [P2] **Jin Hyun Park**, Representations learnt by SGD and Adaptive learning rules: Conditions that vary sparsity and selectivity in neural networks., University of Bristol., 2022., [<https://arxiv.org/abs/2201.11653>]

SELECTED PROJECTS

Leveraging Depth and Attention Mechanisms for Improved RGB Image Inpainting | Python, Pytorch

- [Part of grad. course: *Computational Photography*] Designed a dual-encoder attention architecture that fuses RGB and depth features, enabling depth-aware image inpainting (<https://arxiv.org/pdf/2505.00735>).
- Demonstrated gains over RGB-only baselines across multiple metrics, with Grad-CAM visualizations confirming better spatial reasoning

Personal Photo Search Engine | Python, Pytorch

- [Part of grad. course: *Information Storage and Retrieval*] Implemented image caption generation, query embedding, and advanced face recognition using pre-trained LLM/Transformer models (<https://sites.google.com/tamu.edu/pepse/home>).
- Developed a personal photo search engine, integrating face-matching techniques and sentence-based semantic search features

Dartboard Detection Challenge | C/C++, OpenCV, Python

- [Part of undergrad. course: *Image Processing and Computer Vision*] Trained and evaluated the Viola-Jones framework in practice with a custom dataset. Integrated the Hough line transform and Hough circle transform with the Viola-Jones framework to enhance performance (<https://bit.ly/4azO15c>)
- Applied YOLOv3 to the project and successfully minimized false positives

AWARDS & SCHOLARSHIPS

Graduate Research and Teaching Assistant

- Texas A&M University, 2022 - Present

Travel Grants

- Doosan Corp., Consumer Electronics Show (CES), 2026
- Texas A&M University, Neural IPS, 2025
- Samsung Life Insurance & LG Telecom, Internship, 2024
- Hyundai Mobis, Consumer Electronics Show (CES), 2024

Academic Excellence Grants

- University of Bristol: Future Prize Winner Scholarship 2017 & Barry Thomas Scholarship 2018
- Texas A&M University: 2025-26 Fall & Spring and 2026-27 Fall & Spring

Last Update: July 2026